ModelIt!

Teacher's Guide

The Invisible Force That Charges Your Phone

How Electricity and Magnetism Shape Our World

Quick Reference

Grade Level	Grade 7
NGSS Standard	MS-PS2-3, MS-PS2-5
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

© 2026 Discovery Collective. All rights reserved. For teacher use only.

Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

ModelIt!

Performance Expectation: MS-PS2-3, MS-PS2-5

Ask questions about data to determine the factors that affect the strength of electric and magnetic forces. Conduct an investigation and evaluate the experimental design to provide evidence that fields exist between objects exerting forces on each other even though the objects are not in contact.

Three-Dimensional Learning

Science Practice	**Asking Questions and Planning Investigations** — Students ask questions about data and design investigations to determine factors affecting electromagnetic force strength.
Disciplinary Core Idea	**PS2.B Types of Interactions** — Electric and magnetic forces between a pair of objects do not require contact. The sizes of the forces in each situation depend on the properties of the objects and their distances apart.
Crosscutting Concept	**Cause and Effect** — Students identify that specific changes in current and coils CAUSE predictable changes in magnetic field strength and lifting capacity.

CAST Testing Alignment

MULTIPLE CHOICE:

A student wraps 50 turns of wire around an iron nail and connects it to a battery. The electromagnet picks up 10 paper clips. Which change would most likely INCREASE the number of paper clips it can lift?

CONSTRUCTED RESPONSE:

Design an experiment to test how the number of coil wraps affects the strength of an electromagnet. Include your hypothesis, variables, and how you would measure the results.

2 Scoring Rubric

ModelIt!

Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

ModelIt!

For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

Electromagnetism is one of the four fundamental forces of nature and the foundation of modern technology. Every electric motor, generator, speaker, and MRI machine relies on the connection between electricity and magnetism discovered in the 1800s.

Electricity Creates Magnetism

In 1820, Hans Christian Oersted discovered that electric current creates a magnetic field around a wire. This was revolutionary — it showed that electricity and magnetism are not separate forces but aspects of the same phenomenon. Every wire carrying current is a tiny magnet.

How Electromagnets Work

Wrapping current-carrying wire into a coil concentrates the magnetic field. Each loop of wire adds its field to the next, creating a much stronger combined field. Adding an iron core further amplifies the effect because iron is easily magnetized, aligning its internal magnetic domains with the coil's field.

Factors Affecting Strength

Three main factors control electromagnet strength: (1) Current — more current means a stronger field; (2) Number of coils — more wraps concentrate the field; (3) Core material — ferromagnetic materials like iron amplify the field. These factors combine to determine the total electromagnetic force.

Real-World Applications

Electromagnets are everywhere: electric motors (fans, cars, power tools), generators (power plants), speakers (phones, headphones), MRI machines (hospitals), maglev trains, and junkyard cranes. The ability to turn magnetism on and off with a switch makes electromagnets far more versatile than permanent magnets.

4 The LEVER Framework

ModelIt!

About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — Locate the System
Students identify electric current, number of coil wraps, magnetic field strength, and lifting capacity as components of the electromagnetic system.

E — Establish Relationships
Students determine that both current and coils positively affect field strength, which positively affects lifting capacity.

V — Visualize & Model
Students build a model showing how electrical inputs create magnetic force outputs.

E — Evaluate Outcomes
Students run low power, maximum coils, and full power scenarios to test how variables affect strength.

R — Revise & Extend
Students add core material or distance to explore more complex electromagnetic interactions.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a "first draft" model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: "Based on what you just learned, would you add a component? Change a relationship?" Takes 10–15 minutes.
As a "what-if" tool	When students ask questions during class, say: "Great question — go test that in your model!" Students run new scenarios and report back.
For peer discussion	Partners compare models: "Why does your model have this arrow and mine doesn't?" Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience "what-if" challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

ModelIt!

Slide 1: Cover

2 min

Say: *"Your phone charges wirelessly. Junkyard cranes lift cars with invisible force. How?"*

Watch For

Show video of electromagnetic crane in action.

Slide 2: Learning Objectives

3 min

Say: *"Today we're investigating a force that's literally invisible but runs our entire world."*

Watch For

Have students read objectives. Pre-teach 'electromagnet' vs. 'permanent magnet.'

Slide 3: Big Question

3 min

Say: *"You can't see it. Can't smell it. Can't taste it. But it can lift a CAR."*

Watch For

Demonstrate with magnet and paper clips. Ask: How does it work through air?

Slide 4: LEVER Framework

2 min

Say: *"We'll model how electricity creates magnetism and what controls its strength."*

Watch For

Preview LEVER steps.

Slide 5: Activity 1: Components

8 min

Say: *"What inputs control an electromagnet? What outputs can we measure?"*

Watch For

Guide sorting. Discuss why current and coils are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"More current, more coils — what happens to the invisible field?"*

Watch For

Help students identify all-positive relationships in this system.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test: what creates the strongest electromagnet — more coils or more current?"*

Watch For

Students predict first. Compare single vs. combined variable changes.

Slide 8: Discoveries

5 min

Say: *"How does this invisible force work? What did our model reveal?"*

Watch For

Discuss electricity-magnetism connection. Show field lines with iron filings.

Slide 9: STEM Challenge

12 min

Say: *"You're engineers now. Build the strongest electromagnet you can!"*

Watch For

Distribute materials. Groups compete to lift the most paper clips.

6 Answer Key & Expected Responses

ModelIt!

Component Sorting

- **EXTERNAL:** Electric Current, Number of Coil Wraps
- **INTERNAL:** Magnetic Field Strength, Lifting Capacity

Relationships

- **Electric Current to Magnetic Field Strength:** POSITIVE (+) — More electric current flowing through the wire creates a stronger magnetic field around the coil. More moving charges means more magnetic force.
- **Number of Coil Wraps to Magnetic Field Strength:** POSITIVE (+) — Each additional wrap of wire adds its magnetic field to the total. More coils concentrate and amplify the field.
- **Magnetic Field Strength to Lifting Capacity:** POSITIVE (+) — A stronger magnetic field exerts more force on magnetic objects, allowing the electromagnet to pick up and hold more weight.

Key Discoveries

- Electric current creates a magnetic field around a wire — electricity and magnetism are connected
- More coil wraps concentrate the magnetic field, making the electromagnet stronger
- Increasing current increases the force — more electrons flowing means a stronger field
- Electromagnets can be turned on and off, unlike permanent magnets — this makes them incredibly useful

7 Collaborative Learning Protocols

ModelIt!

Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

ModelIt!

Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Magnets are magic"	Magnetism is a fundamental force of nature with well-understood physics. It results from the motion of electric charges at the atomic level. Every magnet works because of electron behavior.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Electricity and magnetism are unrelated"	Electricity and magnetism are two aspects of the SAME force — electromagnetism. Moving charges create magnetic fields, and moving magnets create electric current. They are fundamentally linked.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Bigger magnets are always stronger"	Size alone doesn't determine magnet strength. An electromagnet's strength depends on current, number of coils, and core material — not just physical size. A small, well-designed electromagnet can be stronger than a large one.